

TCS3351 CRYPTOGRAPHY AND DATA SECURITY
Tutorial 7 - Mathematical Background II

1. Distinguish between a prime and a composite integer. Specify the prime and composite integers in the range from 1 to 30 by using Sieve of Eratosthenes method.

Specify

2,3,.....30

Multiple of 2

4,6,8,10,... even number

Multiple of 3

, 6,9,12,15,18,21,24,27,30

Multiple 5

10,15,20,25

Multiple 7

14,21,28

Multiple of 11

22

Multiple 13

26

Multiple 17

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2,3,5,7,11,13,17,19,23,29 (the remaining number that are not listed)

Answer:

A prime is an integer that has exactly 2 factors (divisors), which are 1 and itself. While a composite integer is an integer that has more than 2 factors (divisors).

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	25
26	27	28	29	30

Primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29

2. Determine whether the following integers are primes or composite. Square root method
- a) 271
 - b) 501 not prime
 - c) 831 not prime
 - d) 1479 not prime
 - e) 3513 not prime
 - f) 6487 not prime

Answer:

- a) $\sqrt{271} = 16.4 \dots$ By dividing 271 with all the primes less than 16, that are 2, 3, 5, 7, 11, and 13, we found that they are indivisible to 271. Hence 271 is a prime.
- b) $\sqrt{501} = 22.4 \dots$ By dividing 501 with all the primes less than 22, that are 2, 3, 5, 7, 11, 13, 17, and 19 we found that 3 divides 501, $501 = 3 \times 167$. Hence, 501 is not a prime.
- c) $\sqrt{831} = 28.8 \dots$ We try 2,3,5,7,11,13,17,19,23, : $831 = 3 \times 277$ (not prime)
- d) $\sqrt{1479} = 38.4 \dots$ We try 2,3,5,7,11,13,17,19,23, : $1479 = 3 \times 493$ (not prime)

The remaining questions are left as exercises. They can be solved using a similar approach to the above.

3. Solve the following problem using the approximation method of calculating $\Pi(n)$ (upper limit discovered by Gauss and lower limit discovered by Legendre).
- a) Find the number of primes is between 100,000 and 200,000.
 - b) Find the number of composite integers is between 100,000 and 200,000.
 - c) What is the ratio of primes to composites in the above range.

Answer:

$$i) \frac{100,000}{\ln 100,000} < \pi(100,000) < \frac{100,000}{\ln 100,000 - 1.08366} \Rightarrow 8686 < \pi(100,000) < 9588$$

$$\frac{200,000}{\ln 200,000} < \pi(200,000) < \frac{200,000}{\ln 200,000 - 1.08366} \Rightarrow 16385 < \pi(200,000) < 17982$$

The number of primes between 100,000 and 200,000:

$$7699 < \pi(200,000) - \pi(100,000) < 8394$$

$$16385 - 8686 = 7699,$$

$$17982 - 9588 = 8394$$

- ii) The number of composite integers between 100,000 and 200,000:
= 100,000 – no of primes integers.

$$91606 < \#(\text{composite integers}) < 92301$$

$$92301 + 7699 = 100000 \text{ (numbers)}$$

$$91606 + 8394 = 100000 \text{ (numbers)}$$

- iii) The ratio of primes to composites:

$$\frac{7699}{91606} < \frac{\pi(200,000) - \pi(100,000)}{\#(\text{composite integers})} < \frac{8394}{92301} \Rightarrow 0.084 < \frac{\pi(200,000) - \pi(100,000)}{\#(\text{composite integers})} < 0.091$$

4. Euler's Totient Function.

- a) State the use of Euler's Totient function and the rules that can be used for finding the function.
- b) Using the appropriate rules, find the value of
- | | |
|------------------|------------------------------------|
| i. $\phi(29)$ | iv. $\mathbb{Z}_{37}^*$ |
| ii. $\phi(47)$ | v. $\mathbb{Z}_{80}^*$ |
| iii. $\phi(102)$ | vi. $\mathbb{Z}_{99}^* = \phi(99)$ |

Answer:

- a) Euler's Totient function $\phi(n)$ is used to compute the number of integers that are smaller and relatively prime to a modulus n . Mathematically, if a modulus n is a product of k primes, $n = p_1^{e_1} p_2^{e_2} \dots p_k^{e_k}$, then:

$$\phi(n) = (p_1^{e_1} - p_1^{e_1-1})(p_2^{e_2} - p_2^{e_2-1}) \dots (p_k^{e_k} - p_k^{e_k-1}).$$

- b) i. Since 29 is a prime, then $\phi(29) = (29 - 1) = 28$.
 iii. Since 102 is a composite such that $102 = 2 \times 3 \times 17$. Hence,
 $\phi(102) = \phi(2) \times \phi(3) \times \phi(17) = (1)(2)(16) = \mathbf{32}$
 v. Since 80 is a composite such that $80 = 2^4 \times 5$. Hence,
 $\phi(80) = \phi(2^4) \times \phi(5) = (2^4 - 2^3)(4) = (16 - 8)(4) = \mathbf{32}$

The remaining questions are left as exercises. They can be solved using a similar approach to the above.

5. For each of the follow problem, solve with the help of either Fermat's Little Theorem or Euler's
- | | |
|-----------------------|------------------------|
| a) $15^{18} \bmod 17$ | d) $27^{-1} \bmod 41$ |
| b) $5^{15} \bmod 13$ | e) $12^{-1} \bmod 77$ |
| c) $15^{-1} \bmod 17$ | f) $16^{-1} \bmod 323$ |

Answer:

- a) Since 17 is a prime, we have $\phi(17) = 16$. By Fermat's Little theorem,
 $15^{18} \pmod{17} \equiv 15^{16+2} \equiv 15^{16} \cdot 15^2 \equiv 1 \cdot 15^2 \equiv 225 \equiv \mathbf{4 \pmod{17}}$.
- b) Since 13 is a prime, we have $\phi(13) = 12$. By Fermat's Little theorem,
 $5^{15} \pmod{13} \equiv 5^{12+3} \equiv 5^{12} \cdot 5^3 \equiv 1 \cdot 5^3 \equiv 125 \equiv \mathbf{8 \pmod{13}}$.
- c) By Fermat's Little theorem, $15^{-1} \pmod{17} \equiv 15^{17-2} \equiv 15^{15} \equiv \mathbf{8 \pmod{17}}$.
- d) By Fermat's Little theorem, $27^{-1} \pmod{41} \equiv 27^{41-2} \equiv 27^{39} \equiv \mathbf{38 \pmod{41}}$.
- e) By Euler's Totient function, $\phi(77) = \phi(7) \cdot \phi(11) = 6 \cdot 10 = 60$. Then, we have
 $12^{-1} \equiv 12^{\phi(77)-1} \equiv 12^{59} \pmod{77} \equiv \mathbf{45 \pmod{77}}$.
- f) By Euler's Totient function, $\phi(323) = \phi(17) \cdot \phi(19) = 16 \cdot 18 = 288$. Then, we have
 $16^{-1} \equiv 16^{\phi(323)-1} \equiv 16^{287} \pmod{323} \equiv \mathbf{101 \pmod{323}}$.

6. Test the following integers whether they are a strong pseudoprime using the Miller-Rabin primality test. (Use at maximum 3 small different bases, such as $a = 2, 3, 5, 7$)
- 997
 - 3801
 - 8997

Answer:

- a) Given $n = 997$, then $n - 1 = 996 = 249 \times 2^2$. We have $m = 249$ and $k = 2$.

Using base $a = 2$:

$k = 0$	$T \equiv 2^{249} \pmod{997} \equiv 161$	
$k = 1$	$T \equiv 161^2 \pmod{997} \equiv 996 \equiv -1$	Prime

Using base $a = 5$:

$k = 0$	$T \equiv 5^{249} \pmod{997} \equiv 161$	
$k = 1$	$T \equiv 161^2 \pmod{997} \equiv 996 \equiv -1$	Prime

Using base $a = 7$:

$k = 0$	$T \equiv 7^{249} \pmod{997} \equiv 161$	
$k = 1$	$T \equiv 161^2 \pmod{997} \equiv 996 \equiv -1$	Prime

Hence, 997 is a strong pseudoprime. (In fact, 997 is a prime number)

- b) Given $n = 3801$, then $n - 1 = 3800 = 475 \times 2^3$. We have $m = 475$ and $k = 3$.

Using base $a = 2$:

$k = 0$	$T \equiv 2^{475} \pmod{3801} \equiv 2606$	
$k = 1$	$T \equiv 2606^2 \pmod{3801} \equiv 2650$	
$k = 2$	$T \equiv 2650^2 \pmod{3801} \equiv 2053$	Composite

Hence, 3801 is a composite, with witness $a = 2$.

The remaining questions are left as exercises. They can be solved using a similar approach to the above.

2249 = 1000 1100 1001
 $A = 2249$, $n = 8997$,

Bit	Multiply if $x_i = 1$, $Y = (a * y) \pmod{n}$	Square $A = a * a \pmod{n}$
1	$Y = 2249$	$2249 * 2249 \pmod{8997} =$

7. Solving the following problem using the Chinese Remainder Theorem.

- a) $x \equiv 2 \pmod{7}$ and $x \equiv 3 \pmod{9}$; $a_1 = 2$, $a_2 = 3$
b) Find an integer that has a remainder of 3 when divided by 7 and 13 but is divisible by 12.

Answer:

a) $M = 7 \times 9 = 63$

$M_1 = 9$, then $M_1^{-1} \equiv 9^{-1} \equiv 4 \pmod{7}$

$M_2 = 7$, then $M_2^{-1} \equiv 7^{-1} \equiv 4 \pmod{9}$

$$x = (2 \cdot 9 \cdot 4) + (3 \cdot 7 \cdot 4) = 156 \equiv \mathbf{30 \pmod{63}}.$$

- b) $x \equiv 3 \pmod{7}$; $x \equiv 3 \pmod{13}$; $x \equiv 0 \pmod{12}$; definition.

$M = 7 \times 13 \times 12 = 1092$

$M_1 = 156$, then $M_1^{-1} \equiv 156^{-1} \equiv 2^{-1} \equiv 4 \pmod{7}$

$M_2 = 84$, then $M_2^{-1} \equiv 84^{-1} \equiv 6^{-1} \equiv 11 \pmod{13}$

$M_3 = 91$, then $M_3^{-1} \equiv 91^{-1} \equiv 7^{-1} \equiv 7 \pmod{12}$

$$x = (3 \cdot 156 \cdot 4) + (3 \cdot 84 \cdot 11) + (0 \cdot 91 \cdot 7) = 4644 \equiv \mathbf{276 \pmod{1092}}.$$

8. Solve the following using the square-and-multiply method of fast exponentiation algorithm.

a) $320^{23} \pmod{461}$

c) $1234^{12} \pmod{4099}$

b) $1736^{41} \pmod{2134}$ (do this now)

d) $491^{37} \pmod{501}$ (do this one)

$23 = 16 + 4 + 2 + 1 = 10111$

Bit	Multiply if $x_i = 1$, $Y = (a * y) \pmod{n}$	Square $A = a * a \pmod{n}$
1	$y = 320.1 \pmod{461} = 320$	$a = 320.320 \pmod{461} = 58$
1	$y = 58.320 \pmod{461} = 120$	$A = 58.58 \pmod{461} = 137$
1	$Y = 137.120 \pmod{461} = 305$	$A = 137.137 \pmod{461} = 329$
0	Skip multiplication . $y = 305$	$A = 329.329 \pmod{461} = 367$
1	$Y = 367.305 \pmod{461} = 373$ (answer)	

$41 = (101001)_2$

do this question $1736^{41} \pmod{2134}$

Answer:

a) $23 = (10111)_2$

Bit	Multiply	Square
1	$1 \cdot 320 \equiv 320 \pmod{461}$	$320^2 \equiv 58 \pmod{461}$
1	$320 \cdot 58 \equiv 120 \pmod{461}$	$58^2 \equiv 137 \pmod{461}$
1	$120 \cdot 137 \equiv 305 \pmod{461}$	$137^2 \equiv 329 \pmod{461}$
0		$329^2 \equiv 367 \pmod{461}$
1	$305 \cdot 367 \equiv 373 \pmod{461}$	

Hence, $320^{23} \pmod{461} \equiv \mathbf{373}$.

b) $41 = (101001)_2$ $1736^{41} \pmod{2134}$

Bit	Multiply	Square
1	$1 \cdot 1736 \equiv 1736 \pmod{2134}$	$1736^2 \equiv 488 \pmod{2134}$
0	$Y = 1736$	$488^2 \equiv 1270 \pmod{2134}$
0	$Y=1736$	$1270^2 \equiv 1730 \pmod{2134}$
1	$1736 \cdot 1730 \equiv 742 \pmod{2134}$	$1730^2 \equiv 1032 \pmod{2134}$
0	$Y=742$	$1032^2 \equiv 158 \pmod{2134}$
1	$742 \cdot 158 \equiv 2000 \pmod{2134}$	

Hence, $1736^{41} \pmod{2134} \equiv \mathbf{2000}$.

The remaining questions are left as exercises. They can be solved using a similar approach to the above.